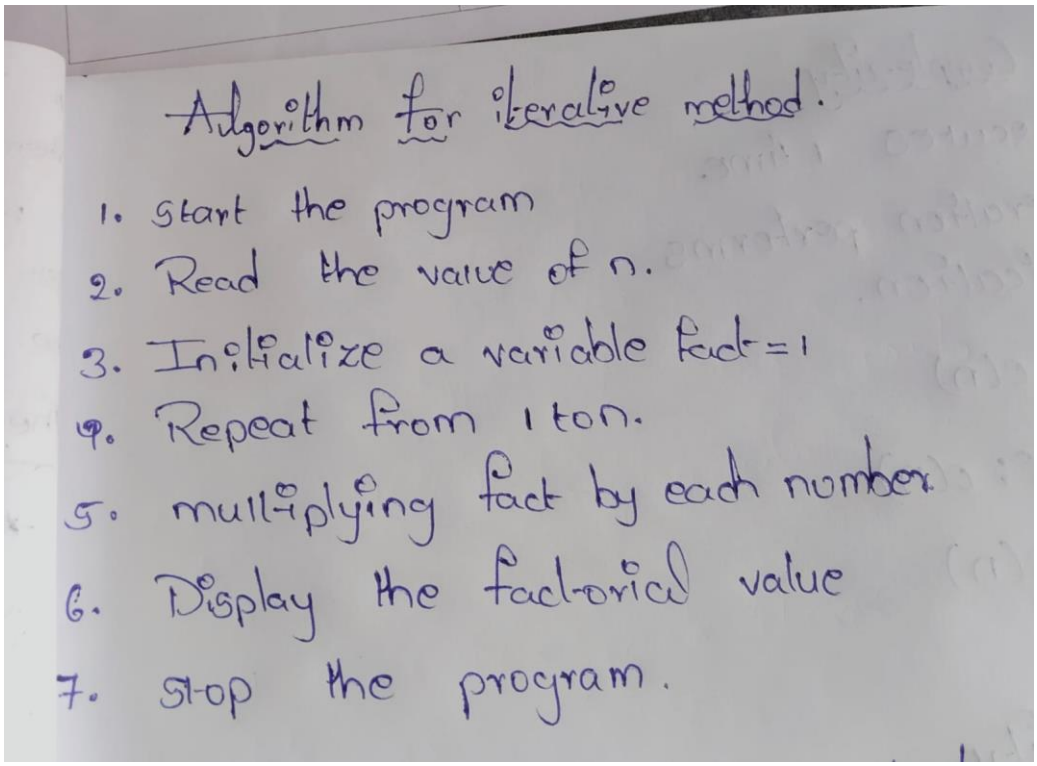
PRACTICAL 4

AIM:

To Implementation and Time analysis of factorial program using iterative and recursive method.

ITERATIVE METHOD

ALGORITHM:



CODE:

```
#include <iostream>

using namespace std;


int main() {

   int n;
```

```
    long long fact = 1;

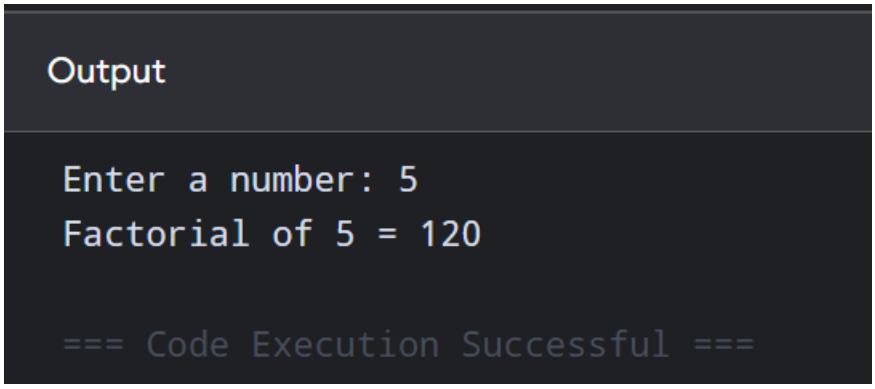
    cout << "Enter a number: ";
    cin >> n;

    for (int i = 1; i <= n; i++) {
        fact = fact * i;
    }

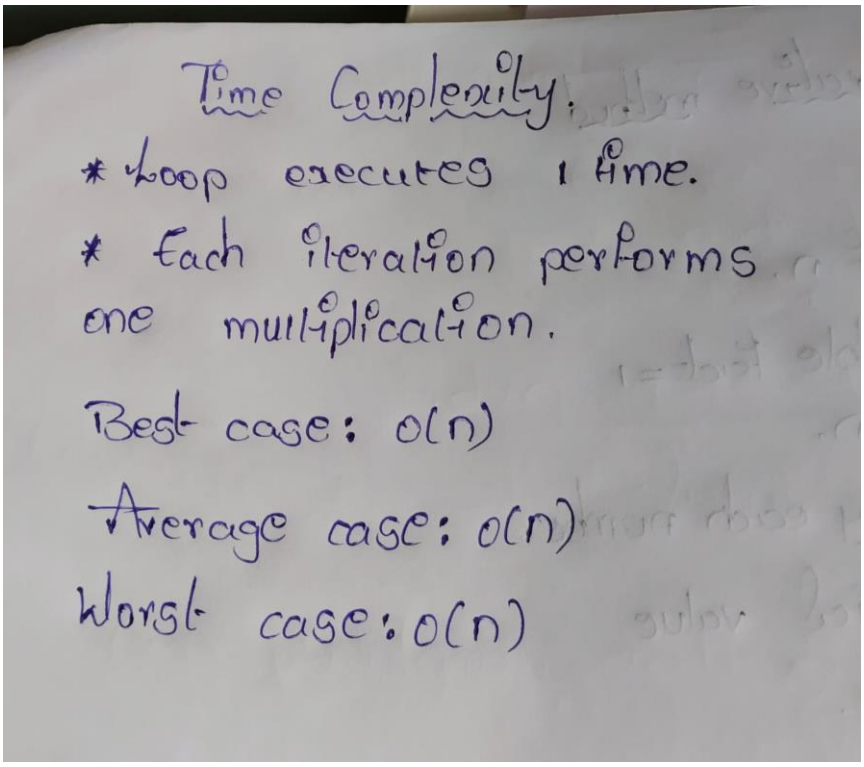
    cout << "Factorial = " << fact << endl;

    return 0;
}
```

OUTPUT:

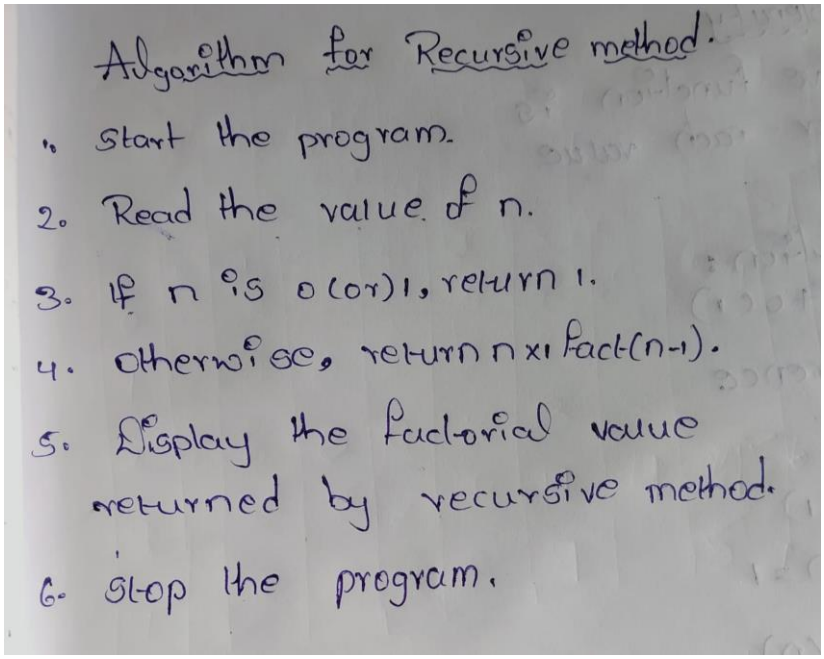


TIME COMPLEXITY:



RECURSIVE METHOD:

ALGORITHM:



CODE:

```
#include <iostream>

using namespace std;


long long factorial(int n) {
    if (n == 0 || n == 1) {
        return 1;
    }
    return n * factorial(n - 1);
}


int main() {
    int n;


    cout << "Enter a number: ";
    cin >> n;


    cout << "Factorial of " << n << " = " << factorial(n);


    return 0;
}
```

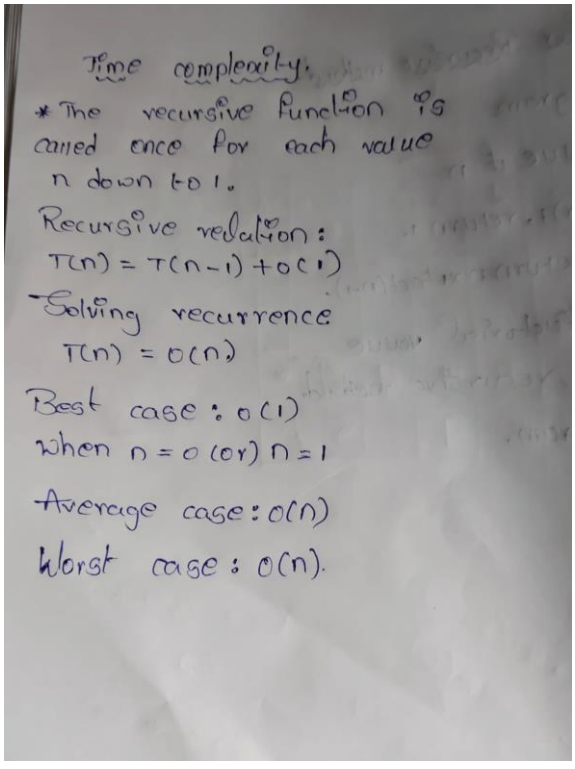
OUTPUT:

```
Output

Enter a number: 6
Factorial of 6 = 720

=== Code Execution Successful ===
```

TIME COMPLEXITY:



CONCLUSION:

Both iterative and recursive methods calculate factorial in O(n) time, but the iterative method uses less memory, while the recursive method is easier to understand and implement.